Vincent Ondeng

Geospatial Data Scientist | Spatial Analytics | Earth Observation | Machine Learning | Disease Modelling

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Profile

Geospatial data scientist with degrees in Geospatial Engineering and Mathematical Sciences. Experience in Earth observation, GIS, multi-source spatial data integration, machine learning on satellite imagery, reproducible geospatial workflows and mathematical modelling of infectious disease. Research interests include spatial epidemiology, environmental determinants of health, climate-related exposures, urban analytics and machine learning for geospatial problems.

Education

African Institute for Mathematical Sciences (AIMS) Kigali, Rwanda

Master of Science in Mathematical Sciences (infectious disease modelling) *Sept. 2024 – Oct. 2025*

- **Thesis:** Modeling the Impact of Case Management Practices in Private Drug Retail Outlets on Malaria Treatment Outcomes (supervised at the Center for Epidemiological Modeling and Analysis; details under Research Experience)
- **Relevant coursework:** Statistical Machine Learning, Big Data Analytics, Database and Data Management, Data Structures and Algorithms
- Full scholarship from the Gates Foundation

University of Nairobi Nairobi, Kenya

Bachelor of Science in Geospatial Engineering *Sept. 2018 – Sept. 2023*

- **Thesis:** Mapping Malaria Distribution in Kisumu County, with an interactive web map of the results built in Folium (Python); Best Final Year Project, Institution of Engineers of Kenya
- **Relevant coursework:** GIS, Remote Sensing, Digital Image Processing, Geospatial Data Infrastructure

Research Experience

M.Sc. Thesis: Malaria Case Management in Private Drug Retail Outlets 2025

AIMS Rwanda and Center for Epidemiological Modeling and Analysis (CEMA) *Nairobi, Kenya*

- **Question:** how case-management practices in private drug outlets, including failure to test, affect malaria treatment outcomes and continued transmission in Kenya.
- **Data:** the National Malaria Control Programme's 2025 nationwide survey of over 400 private drug retail outlets (outlet audits, mystery-shopper visits, staff interviews), used to set case-management parameters; monthly malaria incidence from the Kenya District Health Information System; CHIRPS rainfall data.
- **Model:** extended SEIR model with treatment-pathway compartments for humans and an SEI structure for mosquitoes, solved in R (deSolve); derived the basic reproduction number, analysed equilibrium stability and showed a forward bifurcation at $R_0 = 1$.
- **Fitting:** five parameters, including the amplitude of rainfall-driven seasonal forcing, estimated by least squares, separately for high-, moderate- and low-burden regions and nationally (to capture heterogeneity); model runs parallelised in R.
- **Analysis:** compared four strategies (current practice, universal testing, universal artemether-lumefantrine access, recommended test-and-treat) and ran a global sensitivity analysis with first-order Sobol indices (Saltelli sampling).
- **Results:** test-and-treat improved national recovery rates by about 16.5% and prevented more than 835,000 healthcare re-entries in 2025–2026, with recovery gains up to 24.5% in low- and moderate-burden regions; parameters involving the correct drug explained about 78% of the variance in recovery improvement, testing alone about 8%.

Professional Experience

Geospatial Data Scientist (Consultant) Oct. 2025 – Sept. 2026

United Nations Human Settlements Programme (UN-Habitat) *Nairobi, Kenya*

- Improved and automated existing workflows for computing SDG indicator 11.3.1 using satellite imagery and demographic datasets across UN-Habitat's city portfolio.
- Classified Landsat and Sentinel imagery from 1990 to 2025 using Random Forest, Support Vector Machine and K-means clustering as part of an automated geospatial pipeline for quantifying urban growth in cities across the world.

- Integrated urban spatial datasets such as Google Open Buildings, Microsoft building footprints, World Settlement Footprint and OpenStreetMap to address differences in their coverage across cities.
- Used NDVI to remove vegetated pixels when converting building footprints into built-up rasters at 10 m and 30 m resolution.
- Leveraged Google Open Buildings data and historical satellite imagery in collaboration with the Namibia Statistics Agency (NSA) to estimate building counts across informal settlements in Windhoek between 2015 and 2025.
- Identified and labeled housing typology classes from satellite and street view imagery to train a Convolutional Neural Network for housing stock estimation in cities in Africa and Southeast Asia.
- Performed quality assurance and validation of geospatial processing workflows to ensure realistic urban growth rate values and reproducible outputs across many cities.

M.Sc. Research Fellow May 2025 – Sept. 2025
Center for Epidemiological Modeling and Analysis (CEMA) *Nairobi, Kenya*

- Formulated a deterministic model to investigate the effect of pharmaceutical practices such as failure to perform malaria testing on continued malaria transmission in Kenya.
- Used National Malaria Control Programme pharmacy audit data collected across the country to inform treatment pathway parameters, and fitted transmission parameters to malaria incidence data.
- Quantified that the Ministry of Health Recommended Test and Treat strategy could avert more than 835,000 healthcare system reentries in 2025–2026 if enforced by the relevant authorities.
- Contributed technical modeling skills to the National Malaria Control Programme modeling work-stream as part of an effort to build in-house modeling capacity within the program.
- Presented the model at the National Malaria Control Programme's Malaria Mathematical Modelling Workshop in Naivasha (July 2025).

Research Assistant Apr. 2024 – Sept. 2024
African Population and Health Research Center (APHRC) *Nairobi, Kenya*

- Integrated Sentinel satellite imagery with demographic data from Kenya National Bureau of Statistics to supply spatial indicators such as roof type and building density to support the IDEAMAPS effort of mapping informal settlements in Nairobi.
- Set up KoboToolbox forms for field interviewers collecting water, sanitation and hygiene data.
- Conducted spatial analysis and geoprocessing in ArcGIS, Python, and R to overlay settlement boundaries with hazard layers, identifying priority locations for water points, toilets, and garbage disposal that informed APHRC's recommendations to Nairobi County.
- Developed proposals on flood modeling and urban heat resilience, overlaying hazard exposure layers with settlement and demographic data to inform policy and urban planning.
- Co-facilitated GIS training workshops for 120 Nairobi County environmental officers and 20 APHRC staff, building their capacity to integrate spatial analysis into waste management planning.
- Wrote the step-by-step training manual used after the workshops.

Geospatial Analyst Jun. 2022 – Mar. 2024
Widacom Technologies Limited *Nairobi, Kenya*

- Delivered GIS and spatial analysis for government and private-sector clients, turning requirements into spatial specifications and clear technical outputs.
- Built and maintained spatial data workflows including structuring, querying, and versioning outputs to support repeatable, multi-client project delivery.
- Automated routine geoprocessing and data preparation in Python (GeoPandas/Shapely): cleaning, CRS harmonization, spatial joins, and export packaging.
- Produced reporting-ready maps, summary tables, and technical notes, applying systematic QA/QC that reduced errors and rework.

Projects

Cloud-based Environmental Indices Platform | *Python, FastAPI, STAC, Dask, xarray, Docker* 2026 – Present

- Built an API that takes in a GeoJSON boundary feature and returns NDVI, EVI, SAVI and MNDWI trends.
- Implemented Dask parallel processing to reduce compute time.
- Containerized the API and front-end with Docker Compose for reproducible deployment.
- Added cloud and shadow masking with the Sentinel-2 Scene Classification Layer and corrected reflectance scaling for EVI and SAVI.

Interactive Transport Corridor Dashboard | *Python, Streamlit, Folium* 2026 – Present

- Building an interactive dashboard from Excel transport and trade-corridor data, combining an interactive map with summary statistics.

Graph Neural Network for Informal Settlement Characterisation | *PyTorch, GeoPandas* 2026 – Present

- Developed a GNN to help characterize informal settlements in Luanda, Angola and Dhaka, Bangladesh.
- Compared outputs against an MLP and Random Forest model.
- Used three adjacency definitions for the graph construction: k-nearest neighbor, Delaunay triangulation and distance-threshold.

Geospatial Language Model for Spatial Coding Tasks | *Python, LoRA, pyproj* 2026 – Present

- Fine-tuning a small open language model with LoRA to return code for GeoPandas, PostGIS, CRS and STAC tasks.

Awards, Scholarships and Certificates

MaMod Africa Scholarship funded by the Gates Foundation, AIMS Rwanda: full scholarship for M.Sc. study (2024).

Best Poster Presentation Award, RCMRD International Conference (2024).

Best Final Year Project, Institution of Engineers of Kenya, Geospatial Engineering (2023).

Certificates: Introduction to Research Ethics; Software Engineering Fundamentals.

Technical Skills

Programming: Python (pandas, NumPy, GeoPandas, Shapely, GDAL, Rasterio, xarray, Dask, scikit-learn), R (deSolve), Julia, SQL, Apache Spark, Git and GitHub
Remote sensing: Landsat and Sentinel imagery, Google Earth Engine, STAC, spectral indices (NDVI, EVI, SAVI, MNDWI), cloud masking, image classification, satellite and street-view image annotation, climate and environmental data (CHIRPS)
GIS and web mapping: ArcGIS Pro, QGIS, PostGIS, Folium
Machine learning: Random Forest, SVM, K-means, CNNs (TensorFlow/Keras), graph neural networks (PyTorch), LoRA fine-tuning, supervised and unsupervised methods, feature engineering, model validation, digital image processing
Modelling: Compartmental (SEIR) models, stability and reproduction-number analysis, least-squares fitting, Sobol global sensitivity analysis
Urban analytics: SDG 11.3.1 and urban indicators, informal settlement mapping, urban risk and vulnerability, housing stock and building morphology
Data collection and infrastructure: KoboToolbox, FastAPI, Streamlit, Docker Compose
Communication: technical documentation, training manuals, report writing, presenting to technical and non-specialist audiences
Languages: Kiswahili (native), English (C1)